Bangladesh Army University of Engineering & Technology (BAUET)

Dept. of CSE

Course Code: CSE- 2107
Course Title: Digital Logic Design

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Lecture-1

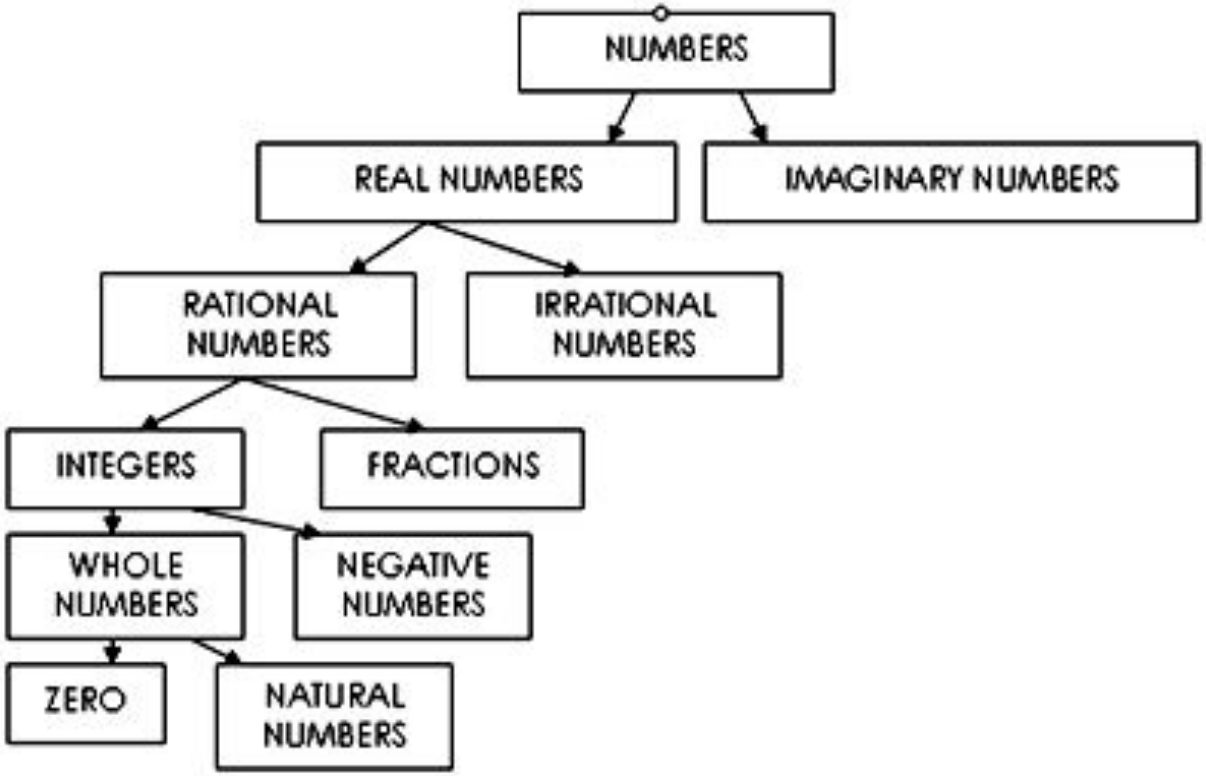
Fundamentals of Digital Logic System

Digital electronics is the foundation of modern computers and **digital** communications.

Massively complex **digital** logic circuits with millions of gates can now be built onto a single integrated circuit (IC) such as a microprocessor.

These IC circuits can perform millions of operations per second.

Number System



Real Number System

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RATIONAL: written in form a/b i.e. fractions and repeating decimals.

INTEGER: whole numbers and their additive inverses. ...-3,-2,-1,0,1,2,3...

WHOLE: 0,1,2,3...

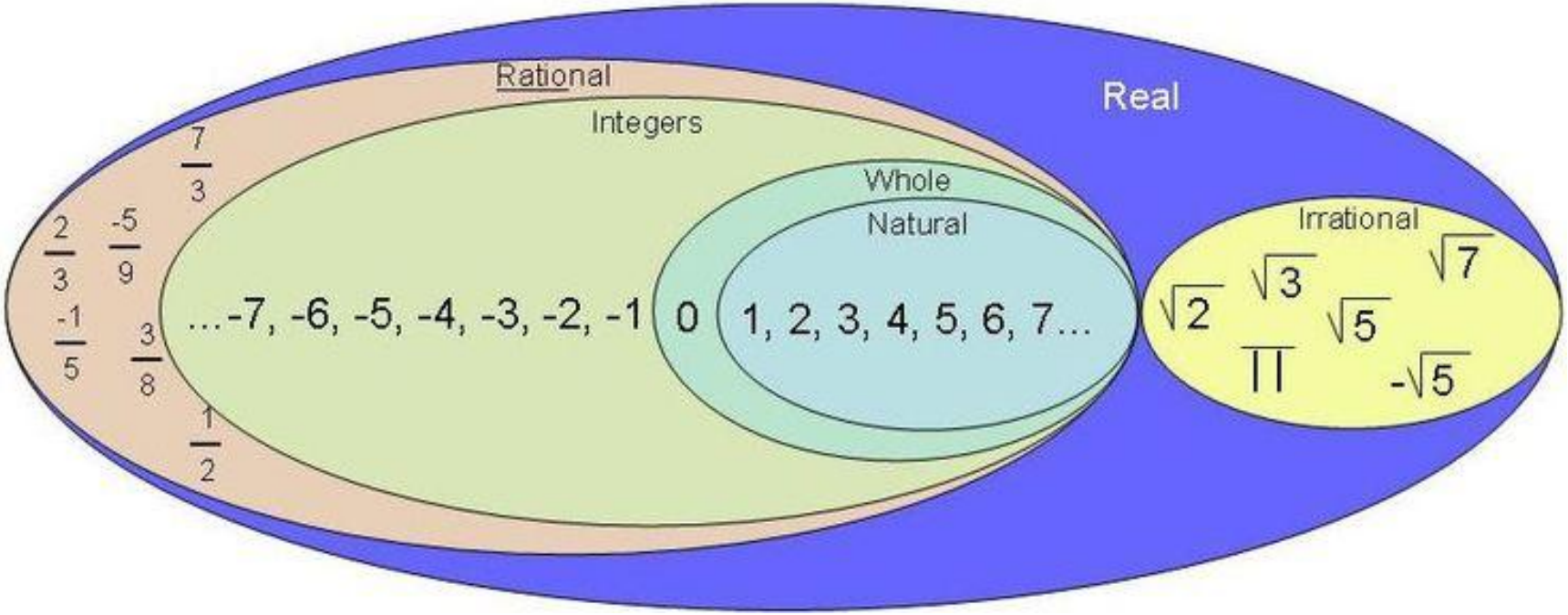
NATURAL (counting): 1,2,3...

REAL: Includes all rational and irrational numbers.

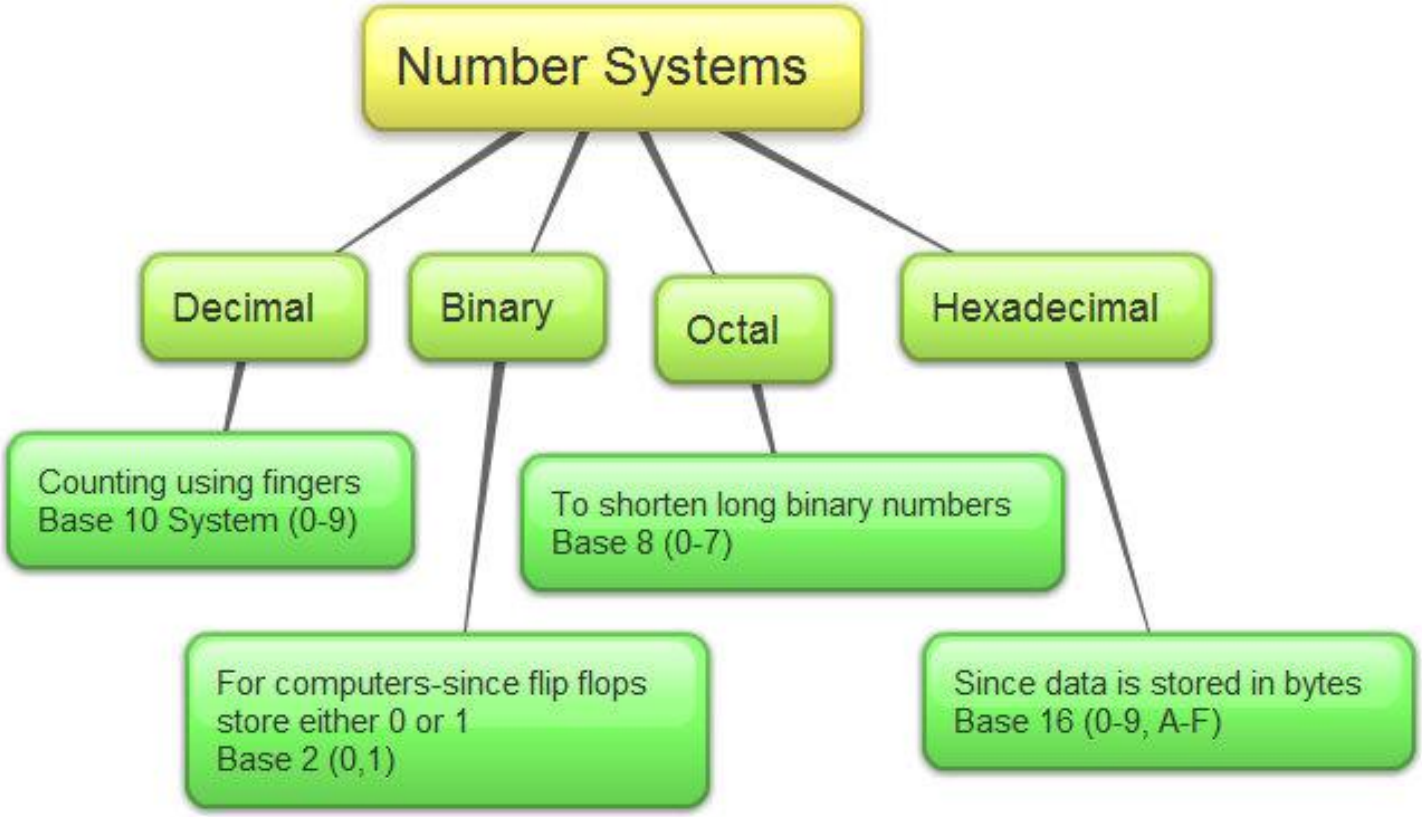
IRRATIONAL: non-terminating, non-repeating numbers, i.e. $\pi = 3.14159...$

www.fingertipmath.com

Real Number System



Common Number Systems



Number Systems

- Each number system is associated with a base or radix

  -The decimal number system is said to be of base or radix 10

- A number in *base r* contains r digits 0,1,2,...,r-1

  -Decimal (Base 10): 0,1,2,3,4,5,6,7,8,9

- Numbers are usually expressed in positional notation

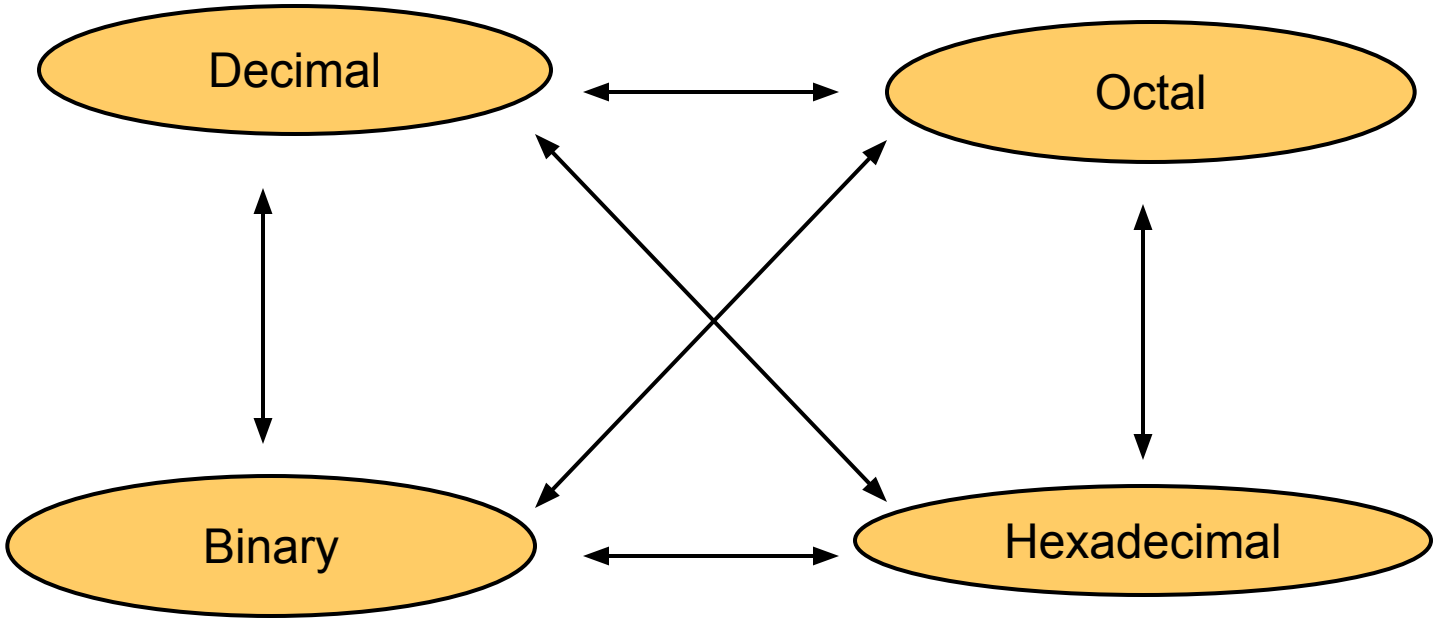
$$N_r = (\underbrace{\overset{\overset{MSD}{\downarrow}}{a_{n-1}} \cdots a_2 a_1 a_0}_{Integer} \overset{\overset{Radix\ Point}{\downarrow}}{\cdot} \underbrace{a_{-1} a_{-2} a_{-3} \cdots \overset{\overset{LSD}{\downarrow}}{a_{-m}}}_{Fraction})_r$$

  - MSD: most significant digit
  - LSD: least significant digit

Common Number Systems

System	Base	Symbols	Used by humans?	Used in computers?
Decimal	10	0, 1, … 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, … 7	No	No
Hexa-decimal	16	0, 1, … 9, A, B, … F	No	No

Conversion Among Bases



Decimal Number

Weight

- 125_{10} => $5 \times 10^0 = 5$

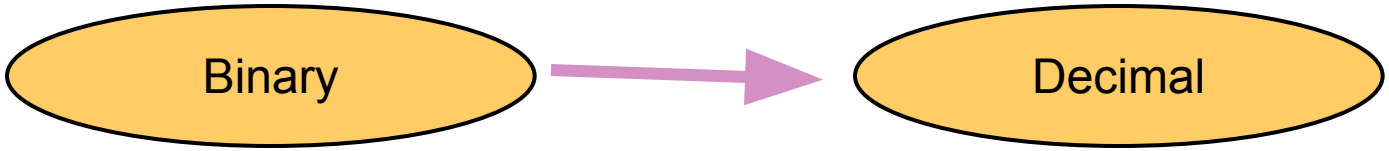
  $2 \times 10^1 = 20$

  $1 \times 10^2 = 100$

  125

Base

Binary to Decimal



Technique

- Multiply each bit by 2^n, where n is the "weight" of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results

Bit "0"

101011_2 =>

$$
\begin{array}{lcl}
1 \times 2^0 & = & 1 \\
1 \times 2^1 & = & 2 \\
0 \times 2^2 & = & 0 \\
1 \times 2^3 & = & 8 \\
0 \times 2^4 & = & 0 \\
1 \times 2^5 & = & 32 \\
\hline
 & & 43_{10}
\end{array}
$$

Decimal to Binary

$125_{10} = ?_2$

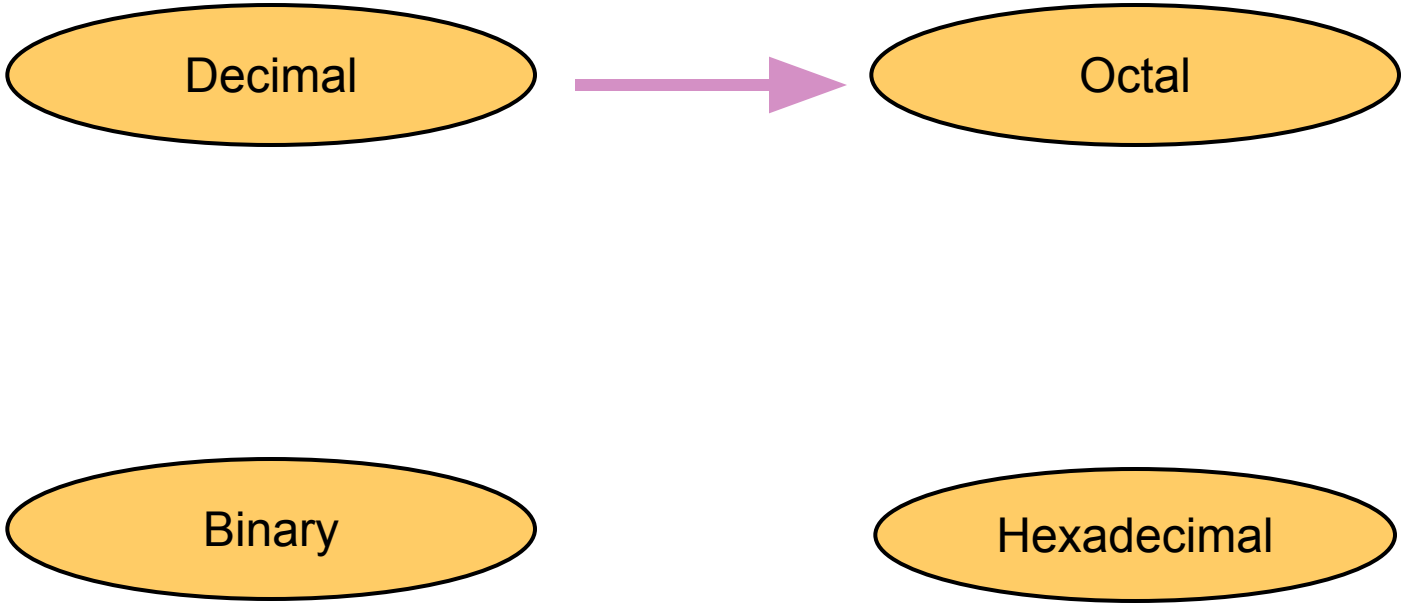
2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

$125_{10} = 1111101_2$

Technique

- Divide by two, keep track of the remainder
- First remainder is bit 0 (LSB, least-significant bit)
- Second remainder is bit 1
- Etc.

Decimal to Octal



Decimal to Octal

- Technique
  - Divide by 8
  - Keep track of the remainder

Example

$1234_{10} = ?_8$

8	1234	
8	154	2
8	19	2
8	2	3
	0	2

$1234_{10} = 2322_8$